DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

FUNCTIONAL REQUIREMENTS & VALIDATION REPORT

TI C2000 F28069M, dual-inverter control, interfaces, and verification

Customer-facing technical report
Revision 1.7 - 5 August 2026

Prepared by	Herder Elektronische Systemen
Primary subject	System requirements, interfaces, verification status, and public release boundary
Evidence basis	Simulink source models, custom ADC source, CAN interfaces, scripts, figures, and schematics
Distribution	Customer-facing; selected source models, custom firmware, and engineering evidence included

DOCUMENT PURPOSE

This report defines what the public Dual-Motor PMSM Dynamometer Platform is required to do, how each requirement is verified, what evidence is retained, and which capabilities remain outside the release. Implementation details and measured-result interpretation are intentionally assigned to the companion engineering report.

1. Executive Summary

The platform is a coupled two-machine dynamometer built around a TI C2000 F28069M controller, two DRV8305 inverter stages, encoder feedback, supervisory host models, CAN communication, and a structured data-processing workflow. This FRD is deliberately concise: it defines mandatory behavior, interfaces, verification methods, and release limitations without duplicating the implementation narrative.

DOCUMENT BOUNDARY

This FRD answers “what shall the system do and how is compliance shown?” The companion Embedded Control, CAN & Characterization Report answers “how was it implemented and what did testing show?”

2. Scope and System Boundary

- Embedded field-oriented control for two coupled PMSMs.
- Drive-machine speed command and load-machine q-axis current command.
- Independent encoder feedback and target-derived acquisition signals.
- Supervisory CAN command path and diagnostic return path.
- Selected editable Simulink models, custom ADC C source, MATLAB utilities, public figures, and interface documentation.

2.1 Excluded claims and assets

- No production functional-safety certification, calibrated torque traceability, EMC qualification, or environmental qualification.
- No generated C/C++ code, binaries, caches, private raw MAT datasets, or workstation-specific build artifacts.
- No vendor manufacturing packages, editable third-party PCB projects, Gerbers, or licensed software bundles.
- No course instructions, student-facing lab material, answer keys, or internal university records.

3. System Context

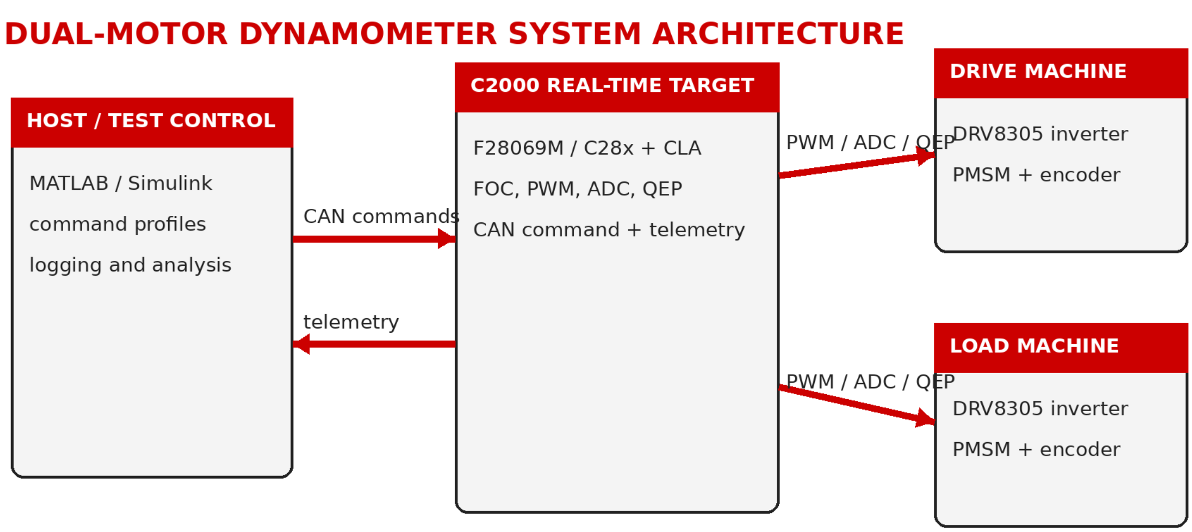

Figure 1. Public system boundary: supervisory host, C2000 real-time target, dual inverter stages, and coupled drive/load machines.

4. Functional Requirements

ID	Area	Requirement	Verification	Status
SYS-001	Real-time control	The system shall execute fast motor-control loops on the embedded target independently of host display timing.	Inspection + functional test	Demonstrated
SYS-002	Dual-machine operation	The system shall command one PMSM as the drive machine and the second as a controllable load or assist machine.	Functional test	Demonstrated
SYS-003	Speed reference	The system shall accept a supervisory speed reference for the drive machine.	Functional test	Demonstrated
SYS-004	Load reference	The system shall accept a q-axis current or equivalent load reference for the load machine.	Functional test	Demonstrated
SYS-005	Enable control	The system shall provide explicit enable and disable control for commanded operation.	Inspection + test	Demonstrated
SYS-006	Feedback	The system shall maintain independent encoder feedback paths for both machines.	Inspection + feedback test	Demonstrated
SYS-007	Observability	The system shall expose speed, current, voltage, torque-related, power-related, and diagnostic quantities required by the public workflow.	Inspection + analysis	Demonstrated
SYS-008	Repeatable workflow	The host workflow shall support repeatable setup, capture, export, and postprocessing.	Procedure review	Demonstrated

5. Interface Requirements

ID	Interface	Requirement	Verification	Status
IF-001	DC input	The inverter stack shall use a verified 24 VDC supply with current limiting during bring-up.	DMM + startup test	Demonstrated
IF-002	Motor phases	Each PMSM shall be connected consistently to its assigned inverter phase outputs.	Continuity + spin test	Demonstrated
IF-003	Encoder routing	Motor 1 and Motor 2 encoder channels shall remain electrically and logically distinct.	Pin inspection + feedback test	Demonstrated
IF-004	CAN physical layer	The CAN bus shall provide CANH, CANL, reference ground, and appropriate termination.	Resistance + bus test	Demonstrated
IF-005	CAN command	The target shall receive the four-word supervisory command payload on the documented command identifier.	Bus test	Demonstrated

ID	Interface	Requirement	Verification	Status
IF-006	CAN diagnostics	The target shall provide a return identifier for command echo or telemetry diagnostics.	Bus test	Demonstrated
IF-007	Custom ADC acquisition	The target shall configure two additional simultaneous ADC input pairs using handwritten F28069M peripheral source.	Source inspection + acquisition test	Demonstrated

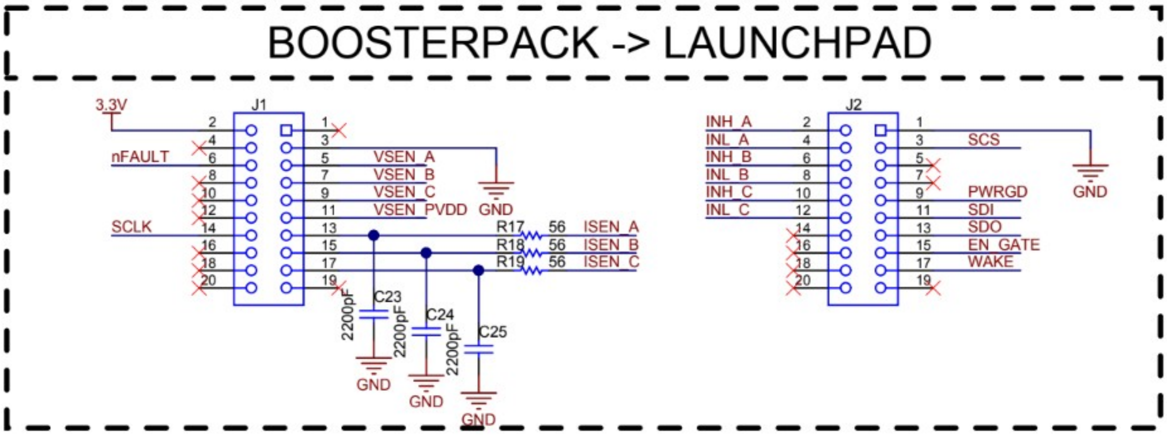

Figure 2. BoosterPack-to-LaunchPad board-interface reference retained with the public release.

6. Data and Processing Requirements

ID	Area	Requirement	Verification	Status
DAT-001	Mechanical power	The analysis shall compute mechanical power from torque and angular speed.	Equation inspection	Verified
DAT-002	Electrical power	The dq-frame workflow shall compute electrical power from 1.5 × Vq × Iq for the retained implementation.	Equation inspection	Verified
DAT-003	Validity filtering	The workflow shall apply explicit minimum-power, minimum-torque, and bounded-efficiency masks before binning.	Script + figure inspection	Demonstrated
DAT-004	Measured bins	Published efficiency maps shall distinguish populated measured bins from unmeasured regions.	Figure inspection	Demonstrated
DAT-005	Sample density	The release shall retain a sample-count view for operating-region confidence review.	Figure inspection	Demonstrated
DAT-006	Traceability	Published figures shall be traceable to retained scripts or approved evidence outputs.	Repository inspection	Verified

7. Safety and Release Constraints

- Verify DC polarity before energizing either inverter stage.
- Begin with low speed, zero load command, conservative limits, and accessible shutdown.
- Confirm encoder direction and phase order before closed-loop sweeps.
- Do not treat software disable as an independent safety function.
- Do not represent the public platform as production-qualified or safety-certified.

8. Requirements Traceability Matrix

Requirement	Verification method	Retained evidence	Status
SYS-001	Inspection + functional test	C2000_Dual_Motor_CAN_Target.slx	Demonstrated
SYS-002	Functional test	target and host models	Demonstrated
SYS-003	Functional test	host setup models	Demonstrated
SYS-004	Functional test	host setup models	Demonstrated
SYS-005	Inspection + test	host/target models	Demonstrated
SYS-006	Inspection + feedback test	board interface + target model	Demonstrated
SYS-007	Inspection + analysis	target model + processing scripts	Demonstrated
SYS-008	Procedure review	setup/logging/analysis scripts	Demonstrated
IF-001	DMM + startup test	interface summary	Demonstrated
IF-002	Continuity + spin test	interface summary	Demonstrated
IF-003	Pin inspection + feedback test	board-interface figure	Demonstrated
IF-004	Resistance + bus test	CAN model set	Demonstrated
IF-005	Bus test	CAN_Receive.slx	Demonstrated
IF-006	Bus test	CAN_Transmit.slx	Demonstrated
IF-007	Source inspection + acquisition test	F28069M_ADC_Custom_Config.c	Demonstrated
DAT-001	Equation inspection	motor_efficiency_processing.m	Verified
DAT-002	Equation inspection	motor_efficiency_processing.m	Verified

Requirement	Verification method	Retained evidence	Status
DAT-003	Script + figure inspection	03_filter_masks.png	Demonstrated
DAT-004	Figure inspection	05_measured_efficiency_map.png	Demonstrated
DAT-005	Figure inspection	06_sample_count_map.png	Demonstrated
DAT-006	Repository inspection	repository map and checksums	Verified

9. Validation and Release Gate

Validation objective	Acceptance basis	Status
Coupled drive/load operation	Motor 1 establishes shaft speed while Motor 2 applies controlled loading.	Demonstrated
Efficiency workflow	Physics masks, measured bins, sample density, and contour outputs retained.	Demonstrated
CAN supervisory control	Command receipt and motor-spin validation retained in public models and documentation.	Demonstrated
Custom ADC path	Additional PWM-triggered simultaneous input pairs integrated with target application.	Demonstrated
Production qualification	Calibration, endurance, EMC, environmental, and independent safety validation.	Not claimed

RELEASE GATE

Publish only after PDFs are rendered page by page, report links resolve, the ZIP opens at repository root, and public files contain no school-specific lab instructions or private build artifacts.